KMTC/QP-08/EPS

# KENYA MEDICAL TRAINING COLLEGE
## FACULTY OF INFORMATION TECHNOLOGY AND MEDICAL EDUCATION
## DEPARTMENT OF HEALTH RECORDS AND INFORMATION

## FINAL QUALIFYING EXAMINATION
## FOR
## DIPLOMA IN HEALTH RECORDS AND INFORMATION TECHNOLOGY

## PAPER: HEALTH STATISTICS

| | |
|---|---|
| **DATE: 15th February 2021** | **TIME: 3 HOURS (9:00AM – 12:00 NOON)** |

**INSTRUCTIONS**

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the EXAMINATION **NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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## SECTION ONE: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. The word statistics is derived from the Latin word
   a) Statistic
   b) Status
   c) Statist
   d) Statista
2. ..... almost always provide better information than do surveys
   a) Data
   b) Experiments
   c) Pilot
   d) Census
3. Data originally collected for an investigation are known as
   a) Data
   b) Primary data
   c) Secondary data
   d) Information
4. Which of the below statement describes secondary data
   a) Should never be used
   b) No scrutiny is required while using it
   c) Should be used after careful scrutiny
   d) While scrutinizing the only thing to see is who collected it
5. When population under investigation is infinite we should use
   a) Sample method
   b) Census method
   c) Sample and census method
   d) Random method
6. The term "errors" in statistics refers to
   a) Mistakes
   b) Bias
   c) Both mistakes and bias
   d) Difference between the value of the statistic and that of corresponding parameter
7. The heading of a row in statistical table is known as
   a) Stub
   b) Title
   c) Caption
   d) heading



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8. Which of the following is not a categorical variable
   a) Numbers of children
   b) Marital status
   c) Political party
   d) Eye colour
9. What of the following should be avoided as a method of presenting data
   a) Spheres and cubes
   b) Bars
   c) Pie diagrams
   d) Pictograms
10. In a moderately skewed distribution the value of mode and median are 8 and 6 respectively. The value of mean in such a situation is approximated equal to
   a) 1
   b) 5
   c) -5
   d) -1
11. When coefficient of skewness is zero, the distribution is
   a) J-shaped
   b) Asymmetrical
   c) Symmetrical
   d) L-shaped
12. If a frequency distribution is positively skewed, the mean of the distribution is:
   a) Less than the mode
   b) Greater than the mode
   c) Equal to the mode
   d) Mean is equal to median
13. When the relationship is of quantitative in nature, the appropriate statistical tool for discovery and measuring the relationship is known as
   a) Regression
   b) Correlation
   c) Skewness
   d) Chisquare
14. The relationship between three or more variables is studied with the help of ..... correlation
   a) Simple
   b) Non-linear
   c) Linear
   d) Multiple



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15. The regression lines cut each other at the point of
    a) Average of X and Y
    b) Average of X only
    c) Average of Y only
    d) Median of X and Y
16. If there is a very strong correlation between two variables then the correlation co-efficient must be
    a) More than +1
    b) Much smaller than 0, if the correlation is negative
    c) Much larger than 0, regardless of whether the correlation is negative or positive
    d) Less than 1
17. If two variables X and Y have a very strong linear relationship, then
    a) There is evidence that X causes a change in Y
    b) There is evidence that Y causes a change in X
    c) There might not be any causal relationship between X and Y
    d) There is a causal relationship between X and Y
18. In regression analysis, the variable that is being predicted is called
    a) Dependent variable
    b) Independent variable
    c) Intervening variable
    d) Predictable variable
19. The classical school of thought on probability assumes that all possible outcomes of an experiment are
    a) Equally likely
    b) Independent
    c) Dependent
    d) Mutually exclusive and equally likely
20. If an event cannot take place, the probability will be
    a) +1
    b) -1
    c) 0
    d) +1 or -1
21. P x Q would always be
    a) Less than one
    b) More than one
    c) Zero
    d) One



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22. Additional theorem states that if two events A and B are mutually exclusive, the probability of occurrence of either A and B is given by

a) P(A) + P(B)
b) P(B) x P(A)
c) P(A) – P(B)
d) P(A) x P(B)

23. If two evens A and B are dependent, the conditional probability of A given B is calculated as

a) P(B/A) (AB)
b) P(B)/P(A)
c) P(AB)/P(A)
d) P(AB)/P(B)

24. The standard deviation of binomial distribution is

a) √nap
b) √npq
c) npq
d) np

25. When variables are not measurable, which of the following is to be represented

a) Nominal
b) Ordinal
c) Interval
d) Ratio

26. Which of the following is not a discrete variate?

a) Skin colour
b) Blood pressure
c) Weight
d) Boys in class

27. Which of the following types of diagrams can be used to find out the relationship between two variables?

a) Pictogram
b) Bar diagram
c) Histogram
d) Scatter diagram

28. A pictorial diagram of frequency distribution is denoted by

a) Line chart
b) Pictogram
c) Histogram
d) Pie chart



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29. The "real limits" of the class interval 2-4 are
    a) 2.5 and 4.5
    b) 2 and 4
    c) 1.5 and 4.5
    d) 1.5 and 3.5
30. If mean, median and mode are 10, 16 and 22 respectively, the distribution is
    a) Symmetric
    b) Normal
    c) Positively skewed
    d) Negatively skewed
31. If "V: stands for variation and "S" stands for standard deviation, then the relation between V and S is?
    a) V=S
    b) $V=\sqrt{S}$
    c) $V^2 = S$
    d) $S^2 = V$
32. The ratio of 3:1 is achieved by
    a) Quartile
    b) Median
    c) 3rd quartile
    d) 1st quartile
33. Area between one standard deviation on either side of mean in normal distribution curve is
    a) 50%
    b) 68%
    c) 136%
    d) 95%
34. A type I error is committed when?
    a) False $H_0$ is accepted
    b) True $H_0$ is rejected
    c) True $H_1$ is accepted
    d) False $H_1$ is rejected
35. P-valve is the probability of
    a) Reflecting $H_0$ when it is true
    b) Accepting $H_0$ when it is true
    c) Rejecting $H_0$ when it is false
    d) Rejecting $H_1$ when it is true



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36. Which of the following is unrelated to the chi-square test of significance
   a) Degree of freedom
   b) Life table
   c) Significance level
   d) Qualitative data

37. Which of the following statements is true about simple random sampling
   a) Techniques that provide least number of possible samples
   b) Every fixed unit is taken for selection
   c) All units have equal chance to be selected
   d) Only selected units have right to be selected

38. For a survey, a village is divided into 5 lanes then each is sampled randomly. It is an example of
   a) Simple random sampling
   b) Stratified random sampling
   c) Systematic random sampling
   d) Multi-stage random sampling

39. Median is almost equivalent to
   a) $25^{th}$ percentile
   b) $50^{th}$ percentile
   c) $75^{th}$ percentile
   d) $10^{th}$ percentile

40. The semi interquartile range is most closely related to
   a) Mean
   b) Median
   c) Mode
   d) Range

## SECTION TWO: SHORT ANSWER QUESTIONS (40 MARKS)

1. Define the following terms as used in probability (5 marks)
   i. Event
   ii. Compound event
   iii. Mutually exclusive events
   iv. Experiment
   v. Trial

2. A binomial distribution has $n = 20$ and $P = 0.3$. Find the mean and standard deviation in this distribution (5 marks)

3. Briefly discuss five (5) disadvantages of interview in data collection. (5 marks)



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4. Compute the coefficient of quartile deviation from the following data (5 marks)

| Marks | Frequency |
|---|---|
| 10 | 4 |
| 20 | 7 |
| 30 | 15 |
| 40 | 8 |
| 50 | 7 |
| 60 | 2 |

5. In the past three months, you have noted that the average length of stay in the facility you are working is 20 days. Advice the management on various actions to be taken to inform their routine decisions (5 marks)

6. In a class of 50 students, 10 have failed and their average marks is 25. The total marks scored by the entire class were 2810. Find the average mark of those who have passed. (5 marks)

7. Before embarking on a data collection exercise, it is imperative to do a pilot survey. Briefly explain how useful the pilot survey is. (5 marks)

8. Briefly describe the four levels of measurement (5 marks)

## SECTION THREE: LONG ANSWER QUESTIONS (20 MARKS)

1. (a) Assuming that a hospital has two departments, past records shows that department 1 collects 30% of the total revenue and department 2 collects 70% of the revenue. Further information indicates that 5% of the revenue collected by department 1 were used for salaries and only 1% collected by department 2 were used for salaries. If a salary is drawn at random, what is the probability that the salary was produced by department 1 or department 2. (10 marks)

   (b) In n experiment on drug testing, Thomas obtained the following frequencies of results: – 315 positives, 101 false positives, 108 negatives and 32 false negatives. According to his theory of testing the numbers should be in proportion 9:3:3:1. Is there evidence to doubt the theory at 5% level of significance? (10 marks)

NB: Attached overleaf find the table for p-values.



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## Significant Values of Chi-square

| | Tail probability $p$ | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| df | .25 | .20 | .15 | .10 | .05 | .025 | .02 | .01 | .005 | .0025 | .001 |
| 1 | 1.32 | 1.64 | 2.07 | 2.71 | 3.84 | 5.02 | 5.41 | 6.63 | 7.88 | 9.14 | 10.83 |
| 2 | 2.77 | 3.22 | 3.79 | 4.61 | 5.99 | 7.38 | 7.82 | 9.21 | 10.60 | 11.98 | 13.82 |
| 3 | 4.11 | 4.64 | 5.32 | 6.25 | 7.81 | 9.35 | 9.84 | 11.34 | 12.84 | 14.32 | 16.27 |
| 4 | 5.39 | 5.99 | 6.74 | 7.78 | 9.49 | 11.14 | 11.67 | 13.28 | 14.86 | 16.42 | 18.47 |
| 5 | 6.63 | 7.29 | 8.12 | 9.24 | 11.07 | 12.83 | 13.39 | 15.09 | 16.75 | 18.39 | 20.51 |
| 6 | 7.84 | 8.56 | 9.45 | 10.64 | 12.59 | 14.45 | 15.03 | 16.81 | 18.55 | 20.25 | 22.46 |
| 7 | 9.04 | 9.80 | 10.75 | 12.02 | 14.07 | 16.01 | 16.62 | 18.48 | 20.28 | 22.04 | 24.32 |
| 8 | 10.22 | 11.03 | 12.03 | 13.36 | 15.51 | 17.53 | 18.17 | 20.09 | 21.95 | 23.77 | 26.12 |
| 9 | 11.39 | 12.24 | 13.29 | 14.68 | 16.92 | 19.02 | 19.68 | 21.67 | 23.59 | 25.46 | 27.88 |
| 10 | 12.55 | 13.44 | 14.53 | 15.99 | 18.31 | 20.48 | 21.16 | 23.21 | 25.19 | 27.11 | 29.59 |
| 11 | 13.70 | 14.63 | 15.77 | 17.28 | 19.68 | 21.92 | 22.62 | 24.72 | 26.76 | 28.73 | 31.26 |
| 12 | 14.85 | 15.81 | 16.99 | 18.55 | 21.03 | 23.34 | 24.05 | 26.22 | 28.30 | 30.32 | 32.91 |
| 13 | 15.98 | 16.98 | 18.20 | 19.81 | 22.36 | 24.74 | 25.47 | 27.69 | 29.82 | 31.88 | 34.53 |
| 14 | 17.12 | 18.15 | 19.41 | 21.06 | 23.68 | 26.12 | 26.87 | 29.14 | 31.32 | 33.43 | 36.12 |
| 15 | 18.25 | 19.31 | 20.60 | 22.31 | 25.00 | 27.49 | 28.26 | 30.58 | 32.80 | 34.95 | 37.70 |
| 16 | 19.37 | 20.47 | 21.79 | 23.54 | 26.30 | 28.85 | 29.63 | 32.00 | 34.27 | 36.46 | 39.25 |
| 17 | 20.49 | 21.61 | 22.98 | 24.77 | 27.59 | 30.19 | 31.00 | 33.41 | 35.72 | 37.95 | 40.79 |
| 18 | 21.60 | 22.76 | 24.16 | 25.99 | 28.87 | 31.53 | 32.35 | 34.81 | 37.16 | 39.42 | 42.31 |
| 19 | 22.72 | 23.90 | 25.33 | 27.20 | 30.14 | 32.85 | 33.69 | 36.19 | 38.58 | 40.88 | 43.82 |
| 20 | 23.83 | 25.04 | 26.50 | 28.41 | 31.41 | 34.17 | 35.02 | 37.57 | 40.00 | 42.34 | 45.31 |
| 21 | 24.93 | 26.17 | 27.66 | 29.62 | 32.67 | 35.48 | 36.34 | 38.93 | 41.40 | 43.78 | 46.80 |
| 22 | 26.04 | 27.30 | 28.82 | 30.81 | 33.92 | 36.78 | 37.66 | 40.29 | 42.80 | 45.20 | 48.27 |
| 23 | 27.14 | 28.43 | 29.98 | 32.01 | 35.17 | 38.08 | 38.97 | 41.64 | 44.18 | 46.62 | 49.73 |
| 24 | 28.24 | 29.55 | 31.13 | 33.20 | 36.42 | 39.36 | 40.27 | 42.98 | 45.56 | 48.03 | 51.18 |
| 25 | 29.34 | 30.68 | 32.28 | 34.38 | 37.65 | 40.65 | 41.57 | 44.31 | 46.93 | 49.44 | 52.62 |
| 26 | 30.43 | 31.79 | 33.43 | 35.56 | 38.89 | 41.92 | 42.86 | 45.64 | 48.29 | 50.83 | 54.05 |
| 27 | 31.53 | 32.91 | 34.57 | 36.74 | 40.11 | 43.19 | 44.14 | 46.96 | 49.64 | 52.22 | 55.48 |
| 28 | 32.62 | 34.03 | 35.71 | 37.92 | 41.34 | 44.46 | 45.42 | 48.28 | 50.99 | 53.59 | 56.89 |
| 29 | 33.71 | 35.14 | 36.85 | 39.09 | 42.56 | 45.72 | 46.69 | 49.59 | 52.34 | 54.97 | 58.30 |
| 30 | 34.80 | 36.25 | 37.99 | 40.26 | 43.77 | 46.98 | 47.96 | 50.89 | 53.67 | 56.33 | 59.70 |
| 40 | 45.62 | 47.27 | 49.24 | 51.81 | 55.76 | 59.34 | 60.44 | 63.69 | 66.77 | 69.70 | 73.40 |
| 50 | 56.33 | 58.16 | 60.35 | 63.17 | 67.50 | 71.42 | 72.61 | 76.15 | 79.49 | 82.66 | 86.66 |
| 60 | 66.98 | 68.97 | 71.34 | 74.40 | 79.08 | 83.30 | 84.58 | 88.38 | 91.95 | 95.34 | 99.61 |
| 80 | 88.13 | 90.41 | 93.11 | 96.58 | 101.9 | 106.6 | 108.1 | 112.3 | 116.3 | 120.1 | 124.8 |
| 100 | 109.1 | 111.7 | 114.7 | 118.5 | 124.3 | 129.6 | 131.1 | 135.8 | 140.2 | 144.3 | 149.4 |

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